



DEPARTMENT OF INFORMATION TECHNOLOGY
LAB MANUAL

CS23432 – Software Construction

(REGULATION 2023)

RAJALAKSHMI ENGINEERING COLLEGE

Thandalam, Chennai-602015

Name: TRINETHRA MAHESH

Register No: 2116241001295

Year / Branch / Section: II / IT / E

Semester: IV

Academic Year: 2025 - 26



RAJALAKSHMI ENGINEERING
COLLEGE
An Autonomous Institution

BONAFIDE CERTIFICATE

Name: TRINETHRA MAHESH

Academic Year: 2025-26 Semester: 1st Branch: IT

Register No

2116241001295

*Certified that this is the bonafide record of work done by the above student in
the CS23432 - Software Construction Laboratory*

during the academic year 2025-2026

Signature of Faculty in-charge

Submitted for the Practical Examination held on 12.05.26

Internal Examiner

External Examiner

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Exp no: 01

Azure DevOps environment setup.

Date: 8.01.26

Aim:

To set up and access the Azure DevOps environment by creating an organization through the Azure portal for the food ordering system.

Procedure:

- i) Open the web browser and navigate to the Azure portal <https://portal.azure.com>
- ii) Sign in using microsoft account credentials. If the user does not have an account create new microsoft account.
- iii) After successful login, the azure home page will be displayed.
- iv) In the search bar, type "Azure DevOps organizations"
- v) Select the Azure DevOps service from the search result
- vi) Click on "My Azure DevOps Organization"
- vii) Click on "Create organization".
- viii) Enter the required details.
 - organisation name
 - Region.
- ix) click continue & complete the setup
- x) after successful creation, the azure DevOps dashboard will be displayed.

Description:

Azure DevOps is a cloud-based platform used to manage software development process such as planning, tracking and collaboration. It helps in organizing the Food ordering system project efficiently.

Result:

~~Successfully~~ accessed Azure DevOps environment & created an organization for food ordering system.

Exp no.: 2

Azure DevOps project setup and user story management.

Date: 22.01.26

Aim:

To setup an Azure DevOps project and manage user stories for the food ordering system.

Procedure:

i) Create Azure DevOps project:

- * Login to Azure DevOps

- * Click on "New project"

- * Enter the project details

Name: Food ordering system

Description: Online system for ordering food.

Visibility: private

- * Click create.

ii) Access project dashboard

After creation, project dashboard will be displayed. It contains boards, repos, pipelines etc.

iii) Create user stories

- * Go to boards → Work items

- * Click on New Work Item → user story

- * Enter title & description.

Sample user stories:

- As a user I want to register, so that I can order food.

- As a user I want to login, so that I can access my account.

- As a user I want to place order, so that I can get food delivered.

- As an admin, I want to manage menu, so that I can update food items.

Description:

Online food ordering & delivery system.

visibility → private

click create

Navigate to boards → work items.

Add user stories using "New Work Item".

User stories define system requirements from the user's perspective and help in agile project development.

Result:

✓ Successfully created project & user stories for food ordering system.

Exp no.: 3

Date: 29.01.26

Setting up epics, features, user stories & tasks.

Aim:

To create epics, features, user stories and tasks for the food ordering system.

Epics:

- i) user management
- ii) Menu management
- iii) Order management
- iv) Payment system
- v) Admin dashboard.

Features:

- ~~User~~ management:
 - registration
 - login
 - profile management.
- Menu management
 - view menu
 - search food items.
- Order management
 - Add to cart
 - Place order
 - cancel order
- Payment system:
 - online payment
 - payment confirmation.

Admin Dashboard

- Manage users
- View reports

User stories

- As a user, I want to login so that I can order food.
- As a user, I want to add items to cart so that I can purchase food.
- As a user, I want to make payment so that order is confirmed.
- As an admin, I want to manage menu so that food items are updated.

Tasks:

1. Design user interface
2. Develop backend API's
3. Database integration
4. Testing & debugging.

Description:

Epics represent major modules, features represent functionalities and user stories represent detailed user requirements.

Result:

~~Successfully~~
tasks.

created epics, features, user stories &

Exp no.: 4

Date: 5.02.26

Sprint Planning.

Aim:

To assign user stories to different sprints for food ordering system.

Sprint planning.

- Sprint 1 (User module)
 - user registration
 - user login.
- Sprint 2 (Menu module)
 - View menu
 - Search food items.
- Sprint 3 (Order module)
 - Add to cart
 - Place order
 - cancel order.
- Sprint 4 (Payment & admin)
 - Payment integration
 - Admin dashboard.

Steps:

1. Go to Boards - sprints
2. Create a new sprint
3. Assign user stories to sprint
4. Set iteration duration

Description: Sprint planning divides the project into manageable phases, ensuring systematic & timely development

Result:

~~Sprints~~ successfully created and user stories assigned.

Exp no.: 5

Poker Estimation.

Date: 12.02.26

Aim:

To estimate effort of user stories using planning poker technique.

Procedure:

- i) Identify all user stories
- ii) Use fibonacci series (1, 2, 3, 5, 8)
- iii) Assign story points based on complexity
- iv) Discuss with team members
- v) Finalize estimation.

Story Point table:

user story	story Points
user registration	3
login	2
view menu	3
place order	5
payment	8
Admin dashboard	8

Explanation:

Planning poker is an agile estimation technique used to estimate the ~~new~~ effort required for completing user stories based on complexity & risk.

Description:

Poker estimation is an agile technique used to estimate the workload collaboratively by & improve planning accuracy.

Result :

~~Successfully~~ estimated user stories using planning
Poker technique.

Exp no.: 06

Date: 19.02.26

Designing class and sequence diagrams
for food ordering system.

Aim:

To design a class diagram and sequence diagram for the food ordering system.

Class Diagram:

Classes used:

User:

Attributes: user ID, name, email, password.

Method: login(), register(), updateprofile().

Restaurant:

Attributes: restaurant ID, name, location

Method: getMenu().

Cart:

Attribute: cart ID

Methods: addItem(), remove item(), calculate total()

Order:

Attributes: order ID, status, total amount

Method: placeorder()

Menu Item:

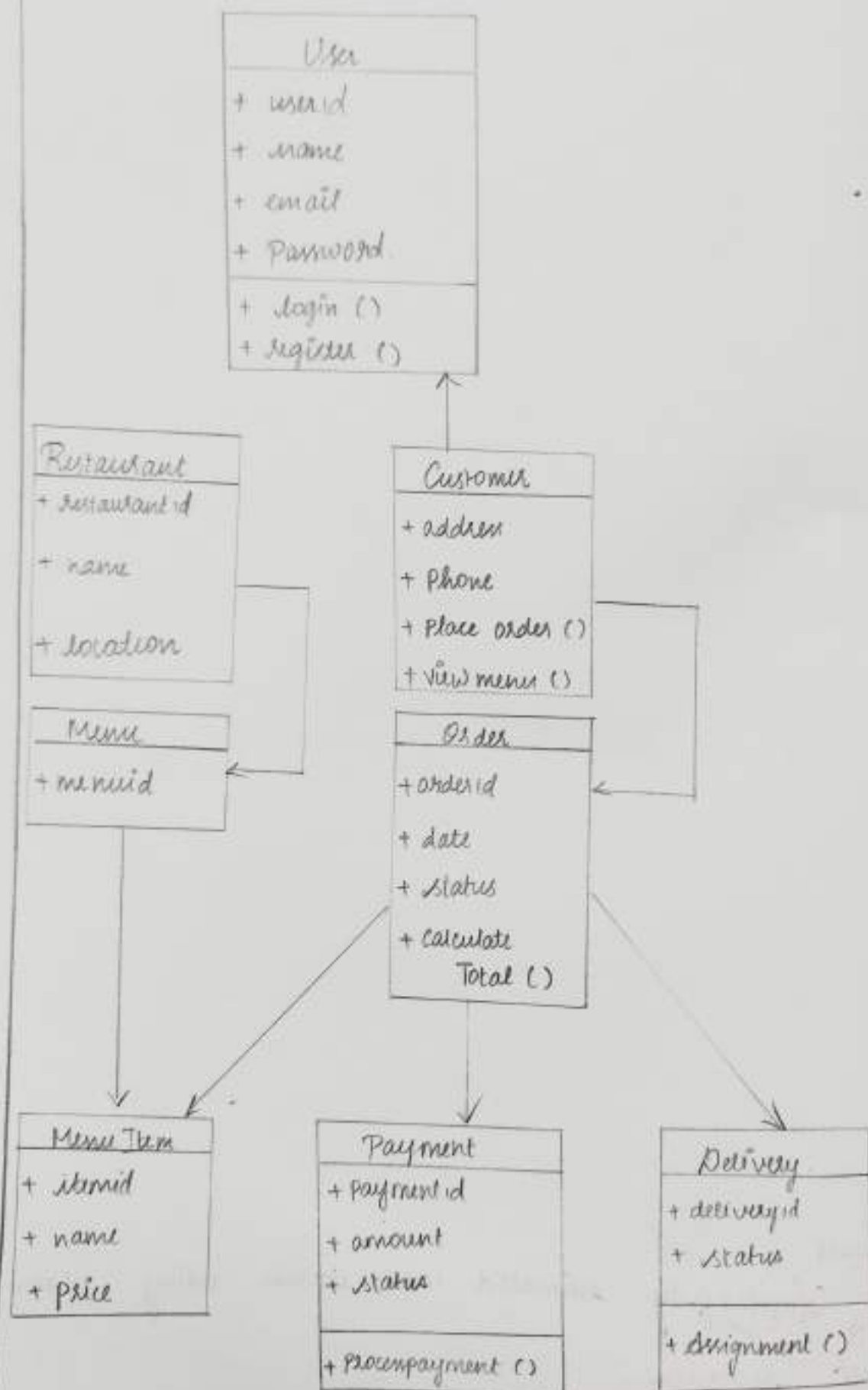
Attributes: itemId, name, price, description

Payment:

Attributes: payment ID, method, status

Method: process payment

Class diagram:



Relationship

- user → cart (user owns a cart)
- user → order (user places orders)
- restaurant → menuItem (restaurant provides menu item)
- cart → menuItem (cart contains items)
- order → menuItem (order includes items)
- order → payment (order is associated with payment)

Sequence Diagram:

Actors & components:

user

App

restaurant

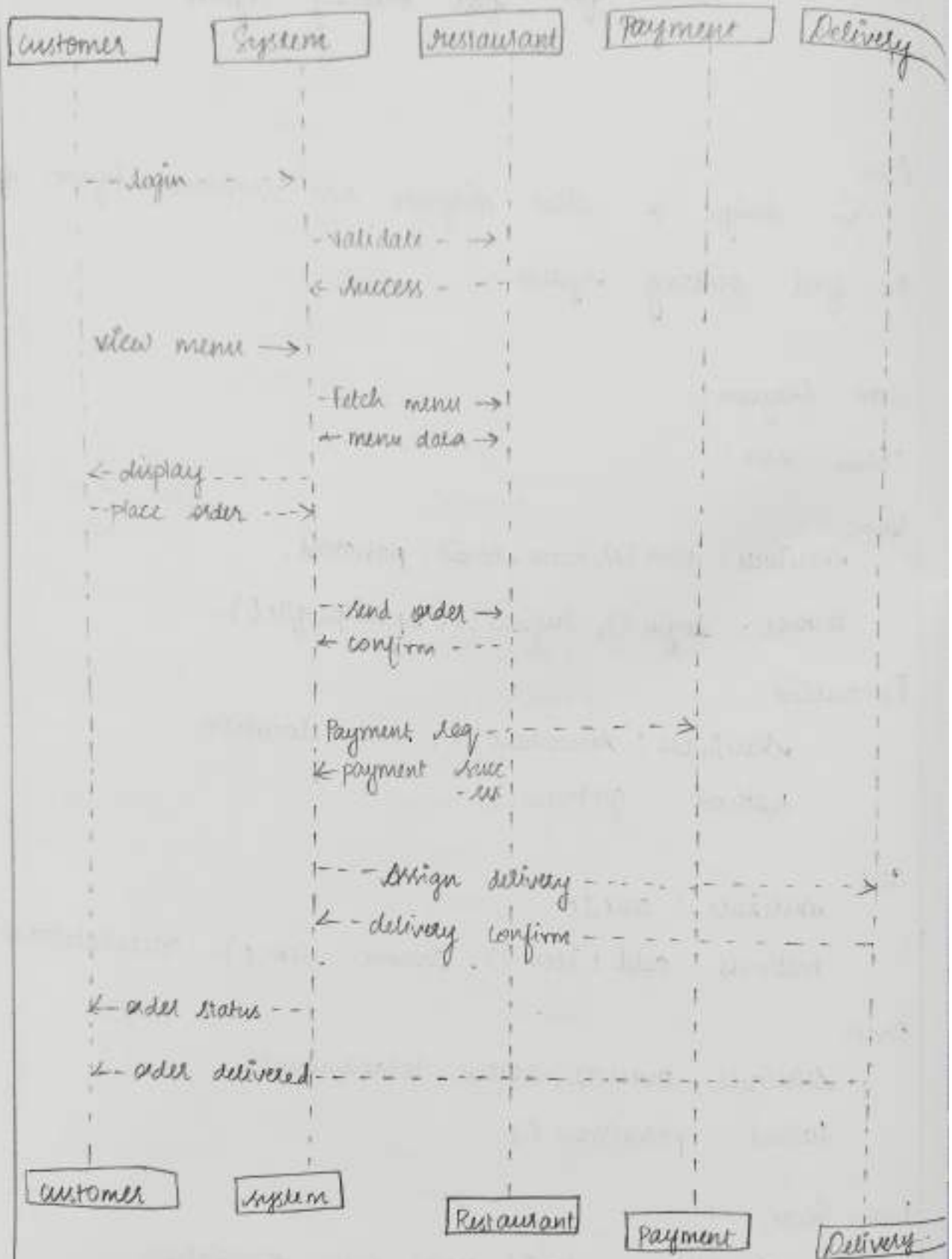
payment gateway

database

Flow of execution:

- i) user logs in or registers through the app
- ii) App verifies user with database
- iii) user browses restaurants
- iv) App fetches menu from restaurant
- v) user adds items
- vi) user places order
- vii) App saves order in database
- viii) payment is processed via payment gateway
- ix) Payment success is returned
- x) Order is sent to restaurant
- xi) Restaurant accepts order
- xii) Order confirm is sent to user.

Sequence Diagram:



Result ~~The~~ class diagram & sequence diagram for the food ordering system are designed successfully.

Exp no : 07

Date : 26.02.26

Designing architectural and ER diagrams for food ordering system.

Aim : To design an architectural diagram & ER diagram for the food ordering system.

Architectural diagram :

Components used :

User : Interacts with the system through the application.

Application (App/Frontend).

Restaurant System

Provides menu details
accepts and places order.

Payment gateway :

Handles secure payment transactions
confirms payment status

Database :

Stores user details, menu items, orders & payment information.

ER Diagram :

entities & attributes

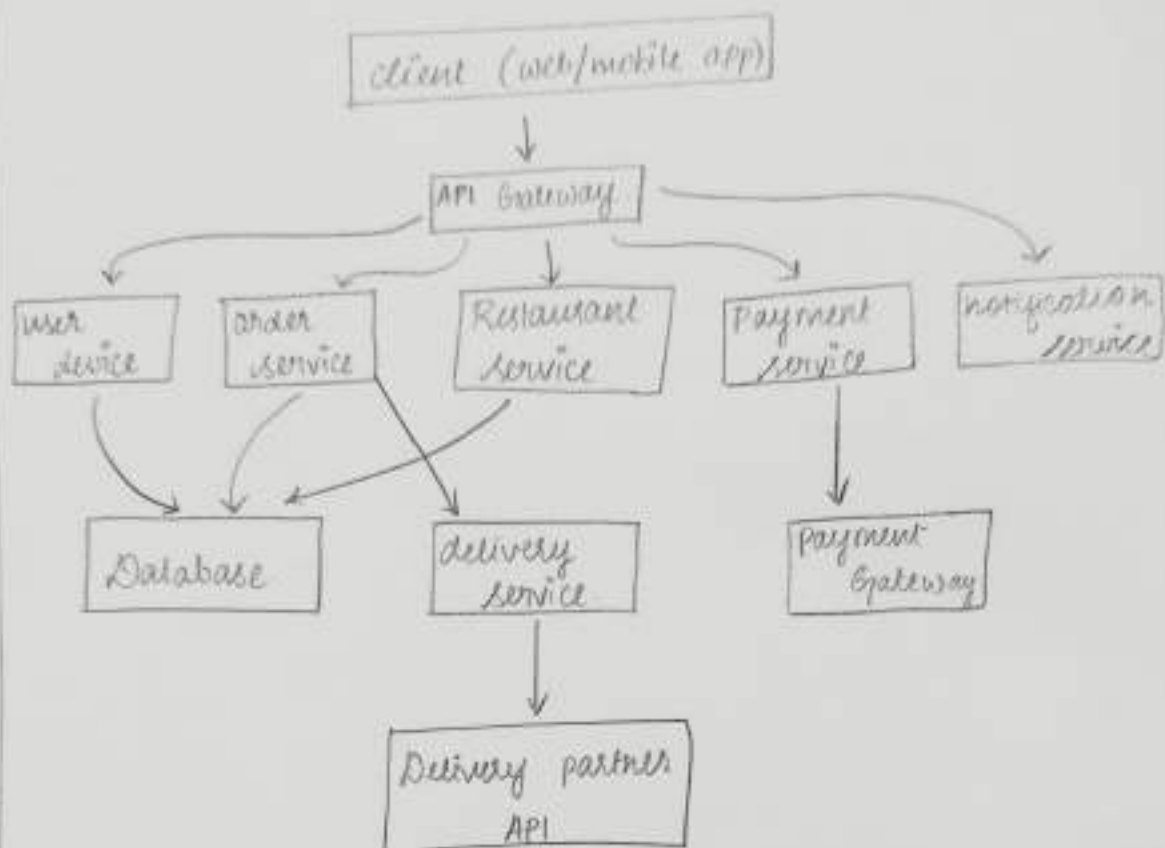
USER :

user_id (primary key)

name :

email :

Architectural Diagram :



Restaurant

restaurant-id (primary key)

name:

location:

menu-item

item-id (primary key)

name

price

Order

order-id (primary key)

order-state

status

orderItem

quantity

Relationships

A user places multiple orders (1:M)

An order contains multiple menuitems (1:M)

A restaurant provides many menuitems (1:M)

order item acts as a linking entity between
order & menuItem.

ER-Diagram:

Restaurant	
int	restaurant-id
String	name
String	location

Customer	
int	cust-id
String	name
String	email
String	phone

Menu Item	
int	item-id
String	name
float	price

Order	
int	order-id
date	order-date
String	status

Order Item	
int	order-id
int	item-id
int	quantity

Payment	
int	payment-id
float	amt
String	status

Delivery	
int	delivery-id
String	status

Result : Thus, the architectural diagram & ER diagram for the food ordering system are designed successfully

Exp no: 8

Date: 5-03-26.

Test plan and test case design for food ordering system.

Aim:

To design the test plan & test cases for validating the food ordering system.

Test plan:

i) Objective:

To ensure that all modules of the food ordering system function correctly & meet user requirements.

ii) Scope:

Testing includes the following modules

User registration & login

Restaurant / menu display

Add to cart

Order placement

Payment processing

iii) Testing type:

Functional testing

Integration testing

System testing.

iv) Test environment:

OS: Windows / Linux.

Browser : Chrome/edge

Database : my sql

Test strategy :

validate user inputs

verify correct order processing

check database updates

Ensure proper UI behaviour.

Test cases :

Test case 1 : user registration.

Test case ID	TC-01
Description	Registers a new user
Input	Name, email, password
Expected output	user registered successfully
Actual output	user register successfully
Status	pass.

Test case 2 : user login

Test case ID	TC-02
Description	Login with valid credentials
Input	Valid email & password
Expected output	Login successful
Actual output	Login successful
Status	Pass.

Test case 3 : View menu :

Test case id	Tc-03
Description	View available food items
Input	Open menu page
Expected output	List of food items displayed.
Actual output	Menu displayed correctly.
Status	Pass.

Test case 4 : Add to cart

Test Card id	Tc-04.
Description	Add food item to cart
Input	Select item & click add
Expected output	Item added to cart
Actual output	Item added successfully
Status	pass.

Test case 5: place order

Test case ID	Tc-05
Description	place food order
Input	confirm cart items
Expected output	order placed successfully
Actual output	order placed successfully
Status	pass.

Result:

The test plan & test cases for the food ordering system were successfully designed & validated

Exp no: 9

Load testing & pipelines.

Date

Aim:

To perform load testing and create a CI/CD pipeline for the food ordering system using azure devops.

Load testing:

Azure load testing is used to evaluate the performance of the food ordering system when multiple users access it simultaneously. It helps in identifying performance issues like slow response time, server overload & failures during peak usage.

Tested APIs:

Place order
View menu
Payment.

Steps:

- Create azure load testing resource
- Add HTTP test (order API end point).
- Configure virtual users (50-100 users)
- Run the test & analyze metrics like, response time & throughput.

Observation:

System performance decreases as users increase, response time is higher during checkout process.

Pipeline:

Pipeline is used to automate the build test &

deployment process. It ensures continuous integration & continuous delivery (CI/CD), reducing manual errors & improving development speed.

Steps:

- Connect Github repository to azure devOps.
- Create azure-pipelines.yml file.
- ⇒ Define pipeline task:
 - Install dependencies
 - Build & run application
 - Execute basic tests.

Advantages:

- Faster deployment
- Automatic error detection
- Continuous monitoring of code changes

Result:

Load testing & pipeline setup for the food ordering system were successfully completed, ensuring better performance reliability & automated deployment.

Exp no: 10

Date:

GitHub Project Structure and Naming.

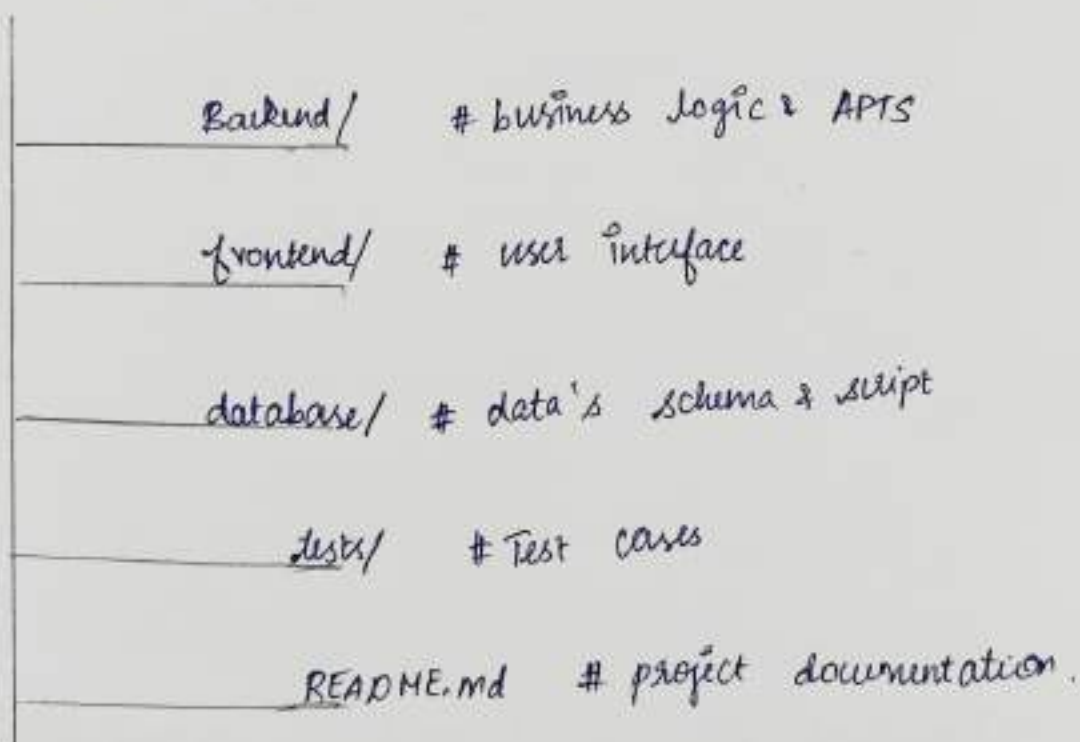
Aim:

To organize the food ordering system project using proper structure & naming conventions.

Project Structure:

A well-defined project structure helps developers easily understand, manage & scale the application.

Food ordering system:



Naming conventions:

Consistent naming conventions improve code reliability & maintainability across the team.

Files → lowercase (order-service.py)
Classes → Pascal case (OrderService)
Variables → camel case (orderId)
APIs → REST format (/api/orders)

Best practices:

- Following best practices ensures clean & efficient development.
- Separate frontend & backend layers
- Use meaningful & descriptive names
- Maintain modules & reusable code
- Keep test files organized separately.

Result:

The project structure is well-organised with proper naming conventions, making the food ordering system easy to develop, maintain & scale.